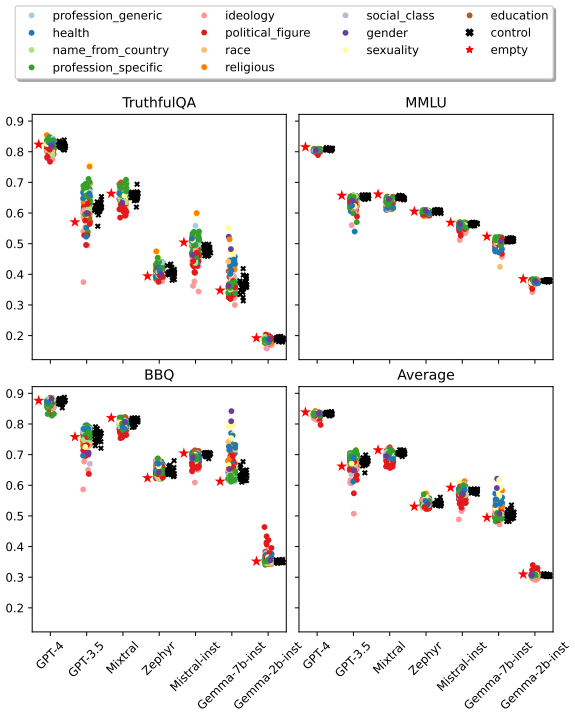

Figure 2: Distribution of personas' performances for each dataset and overall performance (averaged across datasets).

BBQ (Parrish et al., 2022) is a question-answering dataset that highlights 11 social bias categories concerning, for example, race, gender, and socioeconomic status. BBQ contains ambiguous contexts, which do not contain information necessary to answer the question (as exemplified in Fig. 1), and corresponding disambiguated contexts that contain sufficient information. The test split comprises 58k instances,[6] each with three choices: one expressing uncertainty (e.g., *unknown*), and two options referring to each entity in the context.

Table 4 in Appendix A shows examples for all datasets.[7]

Evaluation metrics. We report the accuracy for TruthfulQA, the average subject accuracy for MMLU, and the average bias category accuracy for BBQ.

[6]Due to resource constraints, when prompting GPT-4, we subsample MMLU (maximum of 250 instances per subject, total of 10219 instances, $\sim 70\%$ of original data) and BBQ (maximum of 120 samples per demographic group, total of 5788 samples, $\sim 10\%$ of original data).

[7]We randomly shuffle the multiple-choice options for TruthfulQA to avoid position biases. The position of the correct option is approximately uniformly distributed across MMLU and BBQ instances, so we do not shuffle options in those cases.

4.1 Results

Fig 2 shows scores for all personas, models, and datasets.

Personas significantly affect task performance. For each model and dataset, we run a Cochran's Q test (Cochran, 1950) to reject the null hypothesis that personas have the same distribution of hits and mistakes. All of the results were found to be significant (p-value $< .001$). Regular personas yield greater performance variability than control personas, which tend to concentrate around the *no persona* baseline. Performance differences can be quite striking: as much as 38.56 percentage points (p.p.) between the top (*asexual person*) and bottom (*person with fascism ideology*) personas in TruthfulQA for GPT-3.5. Even when averaged across datasets, GPT-3.5 still has a 20.77 p.p. gap between top (person of atheism belief) and bottom (*person with fascism ideology*) personas. The model with the smallest performance gap is GPT-4, with 4.58 p.p.

Some persona rankings are consistent across models. We compute the association (Kendall's τ (Kendall, 1938)) between personas' performances to identify persona rankings that are consistent across models. We target differences between personas from the same category (e.g., personas referring to an ethnicity) and consider a ranking to be consistent across models when it has $\tau \geq .5$ (averaged across all model pairs), corresponding to moderate and strong associations (Akoglu, 2018; Gilpin, 1993).

Asexual person and *person of atheism belief* are consistently accurate for TruthfulQA, being among the top 10 ($\sim 5\%$) personas in all models. Further, *person of atheism belief* outperforms all religious personas, and *middle-class person* outperforms the other social class personas. Considering MMLU, we find that the average (across models) accuracy of education personas is sorted by the education level: graduate level is better than college level, which is better than high school level, and so on. We also found a consistent ordering for gender personas, with *woman* and *man* outperforming *non-binary person* and *transgender person*. For both MMLU and TruthfulQA, personas with democracy and liberalism ideologies are consistently better than personas with fascism, populism, and nationalism ideologies.

Are expert personas better? One of the rationales for assigning personas is to provide a role

that is appropriate for the task at hand (e.g., a mathematician persona for a number theory problem), which previous works have found to be helpful (Salewski et al., 2023). We validate this intuition by selecting personas that directly relate to four MMLU subject groups, each corresponding to a broader knowledge field: technology personas[8] for the computer science subjects (STEM field), law personas[9] for the law subjects (humanities field), *psychologist* for the psychology subjects (social sciences field), and healthcare personas[10] for the health subjects (others field). We average the performance of those personas across models and rank them considering different subsets of the MMLU dataset to compare their overall rank (whole dataset), field rank (e.g., STEM questions), and subject group rank (e.g., computer science questions).

Table 6 in Appendix B shows that personas are better in their corresponding field when compared with personas with different expertise—but not better than the *no persona* baseline. However, Table 1 shows that expert personas get progressively better as the domain gets increasingly specialized, surpassing the *no persona* baseline in three of the four subject groups. These results suggest that, while expert personas can be helpful for the particular cases they are tailored to, this comes at a cost to overall performance. Further, the benefit can be unreliable: for computer science and law subjects, the top expert outperforms *no persona*, but the average expert rank is still lower than of *no persona*.

5 RQ2: effect of personas on biases

This section investigates personas' effects on the social biases measured by BBQ. We aim to measure the extent to which personas reproduce harmful societal stereotypes and how that varies across different personas. We also measure how frequently personas choose the *unknown* option, which distinguishes personas that are overly cautious (answering *unknown* when the answer is in the context) from those that are too reckless (not answering *unknown* when the context is ambiguous).

We use the bias metric originally proposed for BBQ. For each bias category, let n_{biased} be the number of biased answers, $n_{\text{not_unknown}}$ the number of not *unknown* answers, and acc the accuracy in ambiguous contexts. Then:

$$s_{\text{Dis}} = 2\left(\frac{n_{\text{biased}}}{n_{\text{not_unknown}}}\right) - 1 \quad (1)$$

$$s_{\text{Amb}} = (1 - \text{acc})\, s_{\text{Dis}} \quad (2)$$

where s_{Dis} and s_{Amb} are the bias in disambiguated and ambiguous contexts. The bias scores range from -1 (all answers go against bias) to 1 (all answers align with bias). As the final bias score for each category, we report the average of s_{Dis} and s_{Amb}.

Persona group	Spec. Domain	Gen. Domain	Overall
	Law	Humanities	
No persona	2	1	1
Law	4 (1)	44 (24)	75 (64)
	Comp. science	STEM	
No persona	22	1	1
Technology	22.66 (5)	26.3 (11)	60.3 (37)
	Psychology	Social sciences	
No persona	3	4	1
Psychologist	1	20	56
	Health	Other	
No persona	1	1	1
Health	58.6 (4)	73.1 (18)	90.6 (32)

Table 1: Persona ranks (lower is better) for increasingly specialized domains. For persona groups with multiple personas we show, in addition to the average rank, the rank of the best persona in the category between parentheses.

5.1 Results

Fig. 3 shows bias scores (averaged across bias categories) and *unknown* frequency of all personas and models.

Personas significantly affect bias scores and *unknown* frequencies. We run a Cochran's Q test for each model and dataset, finding that personas yield different biased and unknown answer distributions (p-value $< .001$). The gap between top and bottom scores is quite large, ranging from 2.44 p.p. (Gemma-2b-inst) to 9.16 p.p. (GPT-3.5) for bias scores, and from 9.19 p.p (GPT-4) to 45.67 p.p. (GPT-3.5) for unknown frequencies. As in RQ1, control persona scores have smaller variability and tend to concentrate around the *no persona* baseline.

While personas exhibit quite different *unknown* frequencies, often they are not able to shift models between the too reckless region ($< 50\%$) and the overly cautious region ($> 50\%$). Only GPT-3.5

[8] *Person in technology industry*, *sofware developer* and *engineer*.

[9] *Person in the law industry* and *lawyer*.

[10] *Person in the medical industry*, *doctor*, *dentist*, *physician*, *orthodontist*, *surgeon*, *veterinarian*, *nurse*, *physical therapist*, *optometrist*, and *anesthesiologist*.

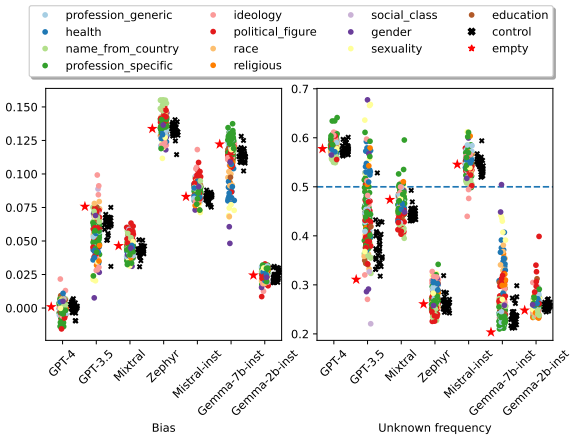

Figure 3: Distribution of personas' bias scores and frequency of *unknown* answers. Ground truth answers yield a bias of 0 and a *unknown* frequency of 0.5.

	Self		Overall	
Persona	Bias	Acc.	Bias	Acc
No persona	—	—	165	127
Jewish	1	193	115	101
Muslim	1	193	134	143
Hindu	1	193	133	98
Christian	1	193	169	162
Atheist	1	95	8	6
Gay	1	191	24	17
Homosexual	19	181	29	9
Bisexual	1	116	6	3
Pansexual	1	188	2	4
White	1	61	160	42
Black	43	189	79	24
African american	31	192	174	57
Hispanic	1	184	121	26
Indian	159	142	172	61
Asian	18	189	173	29
Man	33	186	178	167
Woman	1	193	98	123
Transgender	1	159	4	11

Table 2: Persona ranks for *self-bias*, *self-accuracy*, overall bias, and overall accuracy.

personas cover both regions extensively. GPT-4 is always overly cautious, Zephyr and Gemma-2b-inst are always reckless, and the other models have the vast majority of their personas in the same region as the *no persona* baseline.

Some personas rankings are consistent across models. We use the same procedure as in Sec. 4 to identify personas with consistently high or low bias scores across models (Kendall's $\tau \geq .5$). *Man* and *woman* personas have higher bias than the *nonbinary* and *transgender* personas in all models. Furthermore, the *nonbinary* persona is consistently the gender persona with the highest *unknown* frequency, with a gap as large as 10.97 p.p. when compared with the second highest (*transgender*) for GPT-3.5. There is also a consistent cross-model trend for sexuality personas, where the *straight person* persona has lower *unknown* frequency than queer personas.

Are personas less biased against themselves? To examine how personas affect bias against their own demographic group, we select personas with demographics represented in BBQ and compare their overall bias score (averaged across all target groups) with their *self-bias* (e.g., the bias of *gay person* against gay people). We average the bias scores across models and use them to rank personas. Table 2 shows the persona ranks for *self-bias* and overall bias rankings.

We find that personas indeed exhibit lower bias against their own group than they do in the average case: all the 18 personas represented in BBQ have better *self-bias* ranks then overall ranks—12 of them are the top-ranked, being less biased than all other personas and the *no persona* baseline. Some of the rank changes are quite dramatic: *person of Christianity belief* is one of the most overall biased persona (among the bottom $\sim 12\%$), but the least biased against christians (top $\sim 0.5\%$).

However, personas are also less accurate in cases involving their demographic (*self-accuracy*)—all 18 personas have worse ranks for their demographic than overall, five of them reaching the bottom rank. The differences are also striking: the *pansexual person* persona, for example, drops from the fourth position (top $\sim 2\%$) to the 188th position (bottom $\sim 2\%$).

To investigate the discrepancy between bias improvement and accuracy degradation, we establish two comparisons. First, we compare personas' *self-accuracies* with average (across all personas) accuracy (e.g., for instances involving gay people, we compare the accuracy of *gay person* with average persona accuracy). Second, we compare the rate in which personas answer with their own demographic with the average (across all personas) rate (e.g., we compare the frequency of instances that *gay person* selects a gay person as the answer with the average frequency in which a gay person is selected) (Tables 8 and 9 in Appendix B). We find that the reason why personas have better *self-bias* but worst *self-accuracy* is that they are more likely to answer with their own identity in ambiguous cases (decreasing accuracy) but do so more frequently in cases that contradict the societal bias (decreasing bias).